MANGALAYATAN UNIVERSITY

BESWAN ALIGARH, 202146

SESSION – 2025-26

मङ्गलायतन विश्वविद्यालय

॥ विश्वं ज्ञाने प्रतिष्ठितम् ॥

ASSIGNMENT

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Enrollment No... ; 20251735

Subject Name ; Basic Electrical Engineering

Subject Code ; DEE-201

Class ; Diploma

Semester / Year ; 2nd / 1st

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Sub:- Basic Electrical Engineering
Sub-Code:- DEE-201

Assignment - I

Q1) Explain:

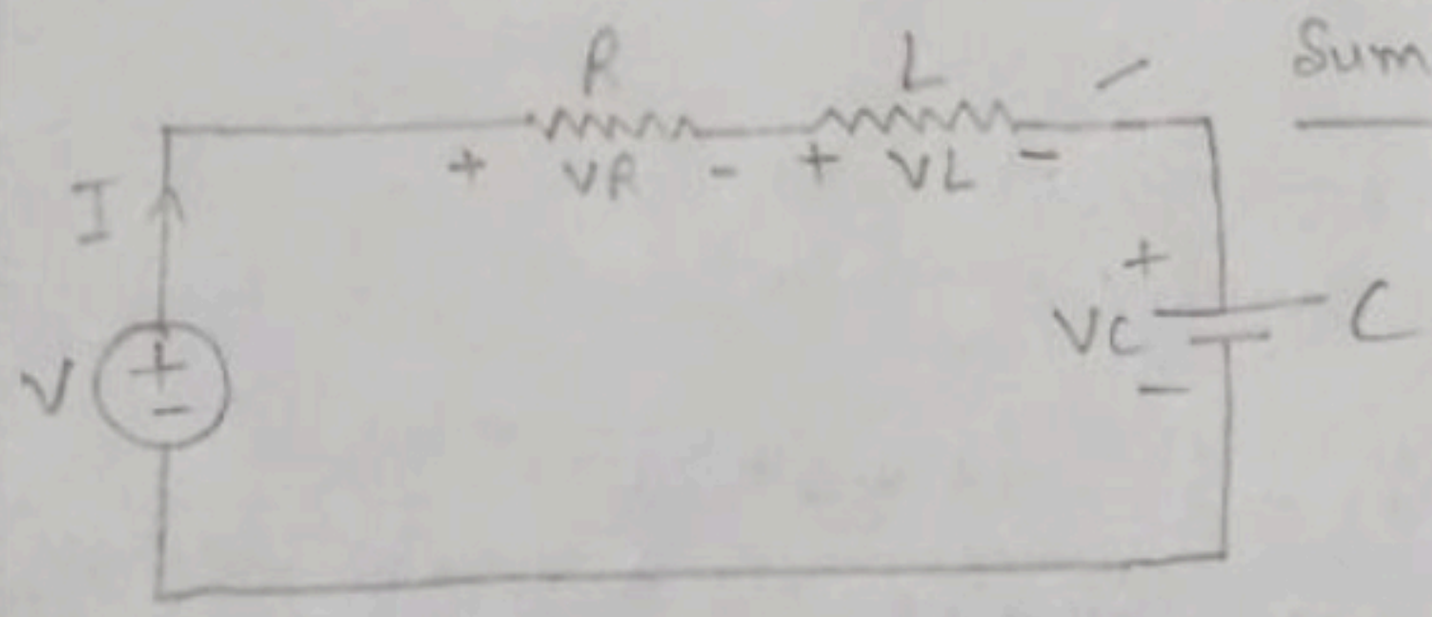
(a) KVL.
(b) KCL

Ans (a) KVL (Kirchhoff Voltage Law)

Statement of KVL:

:- In the Lumped Circuit, the Algebraic Sum of branch voltage around the loop is equal to zero.

In closed loop total sum of voltage zero.



$$-V + V_R + V_L + V_C = 0$$

$$\boxed{V = V_R + V_L + V_C}$$

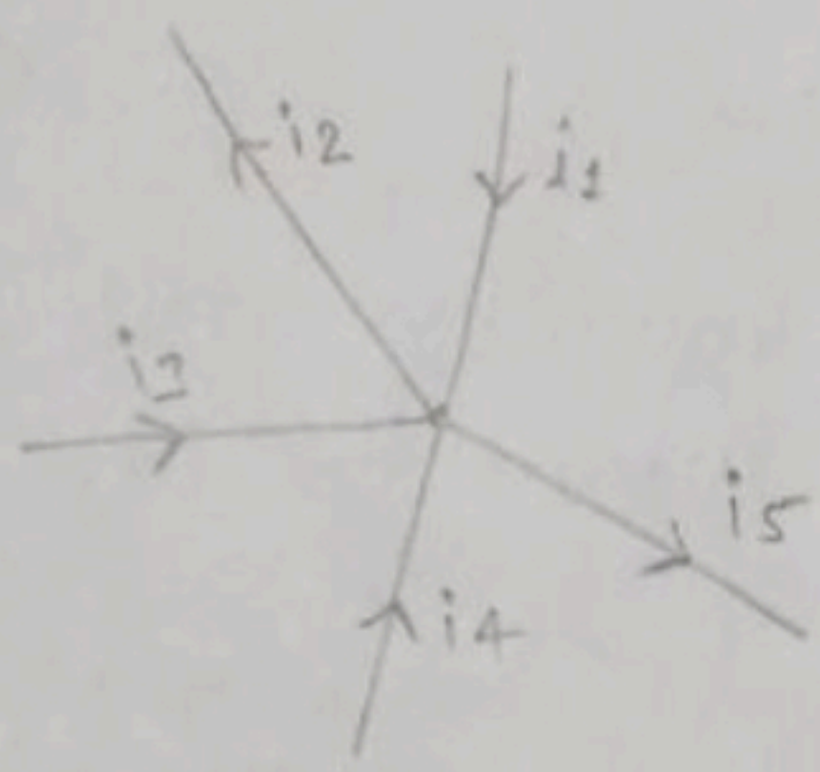
Features of KVL:

(i) Applicable to only lumped circuit.
(ii) Independent of Natural of Element.
(iii) follows: Energy Conservation Principle.

Ans (b) KCL (Kirchhoff Current Law):

Statements of KCL:-

:- In the N/w; the Algebric Sum of the current at a Particular ~~Note~~ Node is equal to zero.



- Incoming current → −ve
- Outgoing current → +ve

$$-i_1 + i_2 - i_3 - i_4 + i_5 = 0$$

$$\boxed{i_1 + i_3 + i_4 = i_2 + i_5}$$ → Sum of Incoming = Sum of outgoing current

Q2) State the Thevenin's theorem and also write down the limitations & applications of the theorem.

Ans The Thevenin's Theorem:

Statement of Thevenin's Theorem:

:- It is a very useful methode of reducing complex linear circuits in to simple circuits, According to this theorem, any complex linear circuit can be reducid to a single voltage source in series with a single resistance.

Different steps for finding thevenin's equivalent are as follows.

(i) open-ckt the terminals of the given circuit with respect to which thevenin equivalent is required.

(ii) The voltage which appears across the open terminals is called. thevenin's voltage Vth.

(iii) Next; voltage sources in the given ckt are shorted and current sources opened. the resistance of the given circuit as viewed from the two open terminals is calculated. at is called thevenin's resistance Rth.

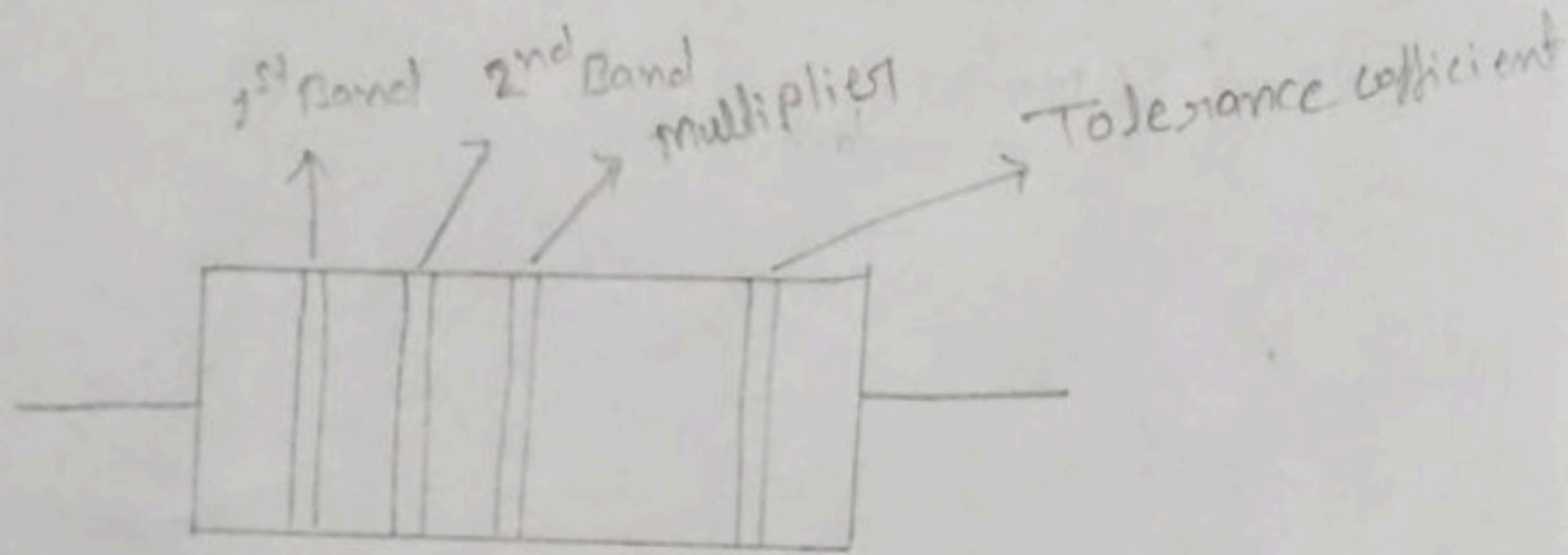
(iv) Voltage Source Vth Connected in Series with Rth gives the thevenin Equivalent for the given network as Shown in fig-A.

$$\boxed{I_L = \frac{V_{th}}{R_{th} + R_L}}$$

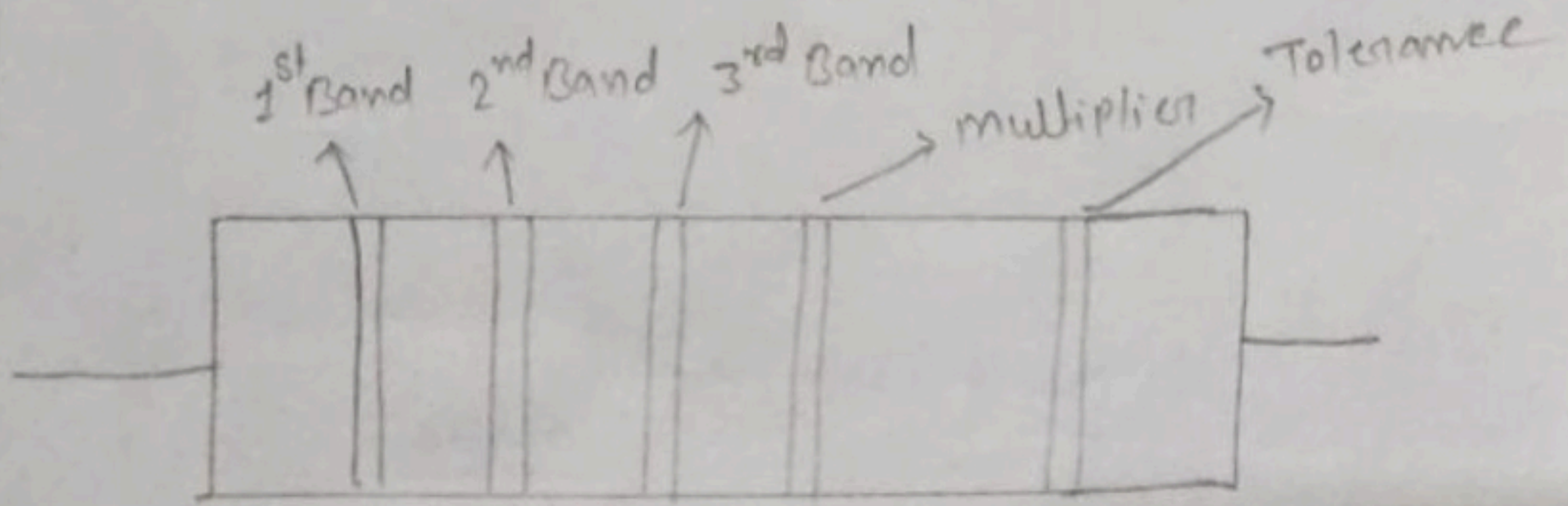
Q3) Explain Colour Coding. Calculate resistance if resistor has the colour band Grey, Black, Blue and gold.

Ans Colour Coding is a System where different Colours are used to represent Specific meanings, Categories, or information so that people can understand Something quickly without reading details.

(i) 4 band Resistor Calculating



(ii) 5- Band Resistor Calculating



Calculate Resistance of resistor has the colour band Grey, Black, Blue and Gold

1st Band – Grey – 8
2nd Band – Black – 0
Multiplier – Blue – 6
Tolerance – Gold – ± 5%

Calculate:

$80 \times 10^6 \pm 5\% \ \Omega$

$\boxed{80M \pm 5\% \ \Omega}$

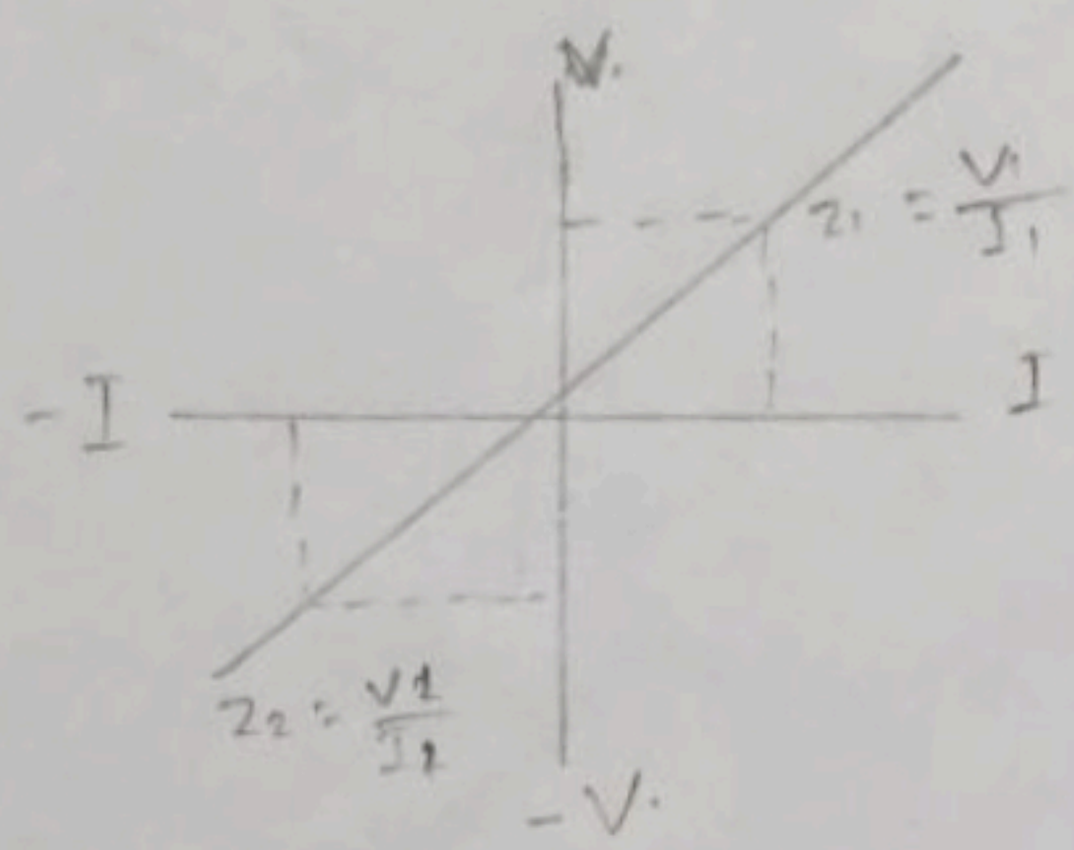
Q4) Write Shorts notes on following . a) Unilateral and Bilateral Element b) Time variant and Time Invariant Element.

Ans a) Unilateral and Bilateral Element:

- Bilateral :- which offeres Same indipendance when the direction / Polarity of voltage/current get reversed other wise unilateral.

Ex. of Bilateral → R, L, C

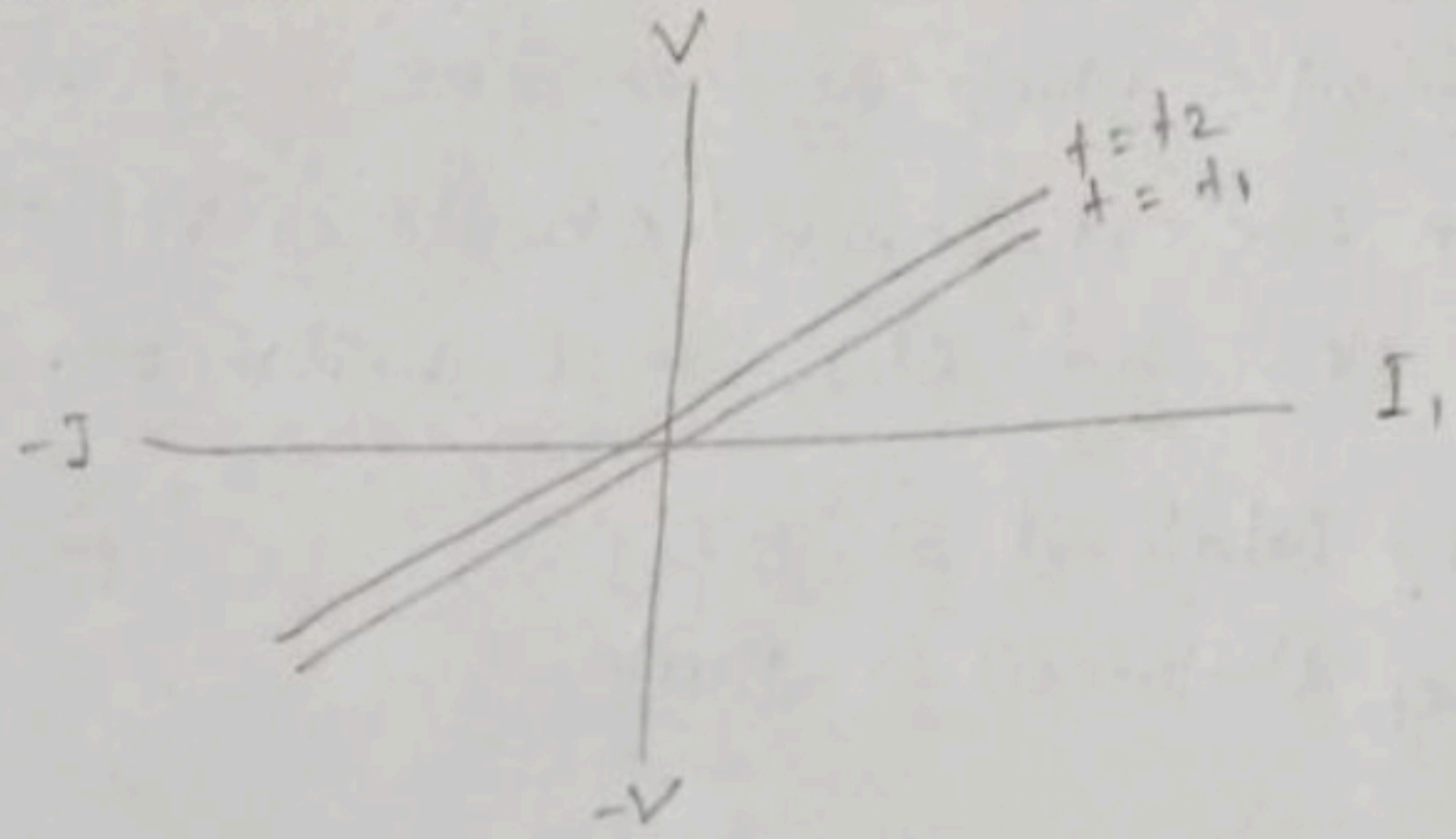
Ex. of Unilateral → Diode



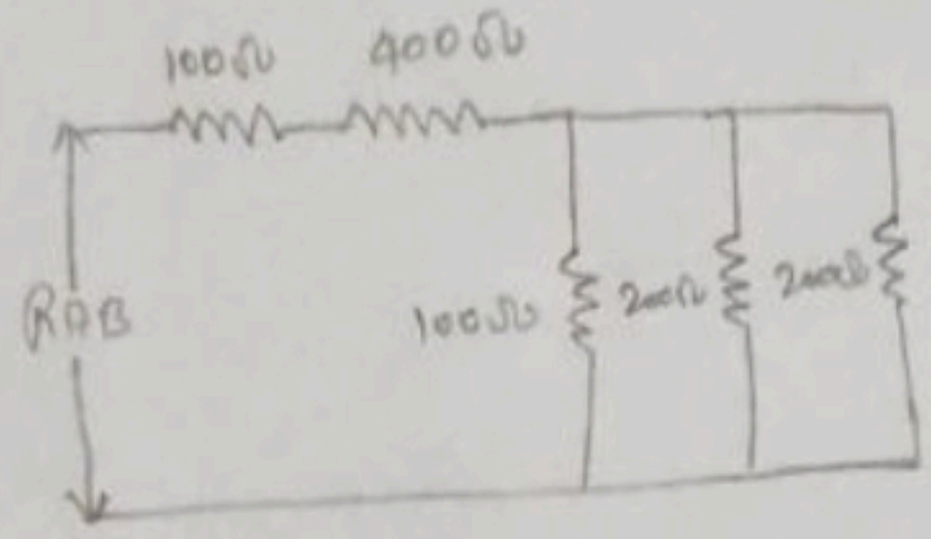
$$Z_2 = \frac{-V_1}{-I_1} = \frac{V_1}{I_1} = Z_1$$

b) Time Variant & Time Invariant:

:- An Element is Said to be time variant where V-I Characteristics changes with time otherwise Time Invariant.



Q5). Calculate the resistance Rab in figure 1.

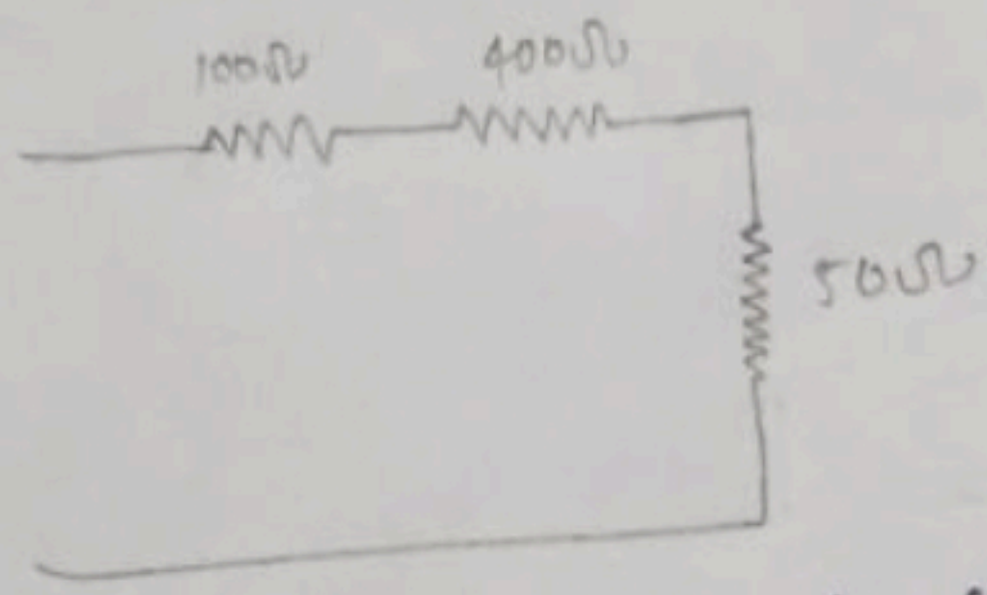


Ans Solve:- 200||200 in Parallel

$$\frac{200 \times 200}{200+200} = \frac{\cancel{40000}^{100}}{\cancel{400}} = 100\Omega$$

100||100 in Parallel

$$\frac{100 \times 100}{100+100} = \frac{\cancel{10000}^{50}}{\cancel{200}} = 50\Omega$$



100Ω, 400Ω, 50Ω in Series

$R_{AB} = 100 + 400 + 50$

$R_{AB} = 550\Omega$